

SI - Forecast Uncertainty Deviation

After the float gets its last Gps position, we do not know its exact future position

So,

Time after Gps fix $\rightarrow$ position Uncertainty

we considered the Uncertainty as a diffusion / random - walk process

for a random process

Variance $\propto$ Time

Therefore

$$\sigma^2(t) \propto t$$

and since standard deviation is the square root of variance

$$\sigma(t) \propto \sqrt{t}$$

At $t=0$, the float already has some Gps/Fix Uncertainty

$$\sigma(0) = \sigma_0$$

We also introduced τ , the current error decorrelation time, to represent how quickly the current error changes

$$\sigma(t) = \sigma_0 \sqrt{1 + \frac{t}{\tau}}$$

$$\frac{t}{\tau}$$

$\sigma_0 \rightarrow$ uncertainty immediately after Gps fix
 $t \rightarrow$ time since last Gps fix
 $\tau \rightarrow$ current error decorrelation time

d/p

$\sigma(t)$ = predicted position uncertainty at time t .

S4 - up - current Loiter point Derivation

The float is moving with the down current.
Suppose the float is predicted to surface at

But the ^{up} guide reaches the waiting location before the float surfaces

so if the guide simply waits at μ_f , the float may move down stream before the actual meeting

Calculate how far the float moves

Current velocity $\boxed{u}$

waiting time $\boxed{t_{wait}}$

Basic motion equation

Distance = Velocity $\times$ Time

$$\therefore d_{drift} = u t_{wait}$$

Because the float will move in the direction of the current, the guide should wait opposite to the current direction

$$\therefore x_{loiter} = \mu_f - u t_{wait}$$

Final formula is

$$\boxed{x_{loiter} = \mu_f - u t_{wait}}$$

Inputs

$\mu_f \rightarrow$ predicted float surfacing position

$u \rightarrow$ ocean - current velocity

$t_{wait} \rightarrow$ expected waiting / transmission time

output

$x_{loiter} =$ position where the glider should wait

57 - polar solar energy distribution

At polar latitudes the amount of sunlight changes dramatically with the season

First calculate solar declination

we use

$$\delta = 23.44^\circ \cos \left[\frac{360^\circ}{365} (N+10) \right]$$

N = day of the year

δ = solar declination

Calculate the Sunset hour angle

At latitude ϕ :

$$\cos \omega = -\tan \phi \tan \delta$$

for Therefore

$$\omega = \cos^{-1} (-\tan \phi \tan \delta)$$

for the project

$$\phi = -62^\circ$$

Convert angle into daylight hours
earth rotates

$$15^\circ / \text{hour}$$

$$\therefore D = 2\omega / 15^\circ$$

Compare energy generated with energy consumed

System load:

E_{load}

$$\therefore NER = \frac{E_{harvest}}{E_{load}}$$

NER = Net energy Ratio

if $NER \geq 1$

then

Energy generated $\geq$ energy consumed

→ system can sustain itself

if

$$NER < 1$$

then

Energy generated $<$ energy consumed

→ battery reserves are required.